Zhelin Sheng

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My Github
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Education

University of Rochester — Rochester, NY
B.S. in Applied Mathematics, B.A. in Computer Science — Aug 2020 – May 2024
- GPA: 3.88

University of Rochester — Rochester, NY
Ph.D. in Statistics — Aug 2024 – Present

Research and teaching experience

VC-dimension and neural networks — Sept 2021 – 2022
University of Rochester, Math department — Rochester, NY
- Assisted in the development of a new tool, discrete s-energy that identifies the salient underlying fractal features of the data set.
- Implemented the new tool outperforming PCA in accessing fractional dimensional phenomena in large data sets with Python.

Neural networks, approximation, and geometric measure theory — 2022 – 2023
University of Rochester, Data Science department, NSF Tripods REU/STEM for All — Rochester, NY
- Applied neural network models on recognizing fractal behavior.
- Provided a deliberate construction of n-dimensional cantor sets which can be reduced to dimension one, with code implementation on testing the neural network's ability to retrieve the original data.
- Explored several quantities that potentially determine the quality of neural network's prediction on time series data.

Fractal structures in large data sets — 2023 – Present
University of Rochester, Data Science department, NSF Tripods REU/STEM for All — Rochester, NY
- Extended and validated the discrete s-energy's function in identifying the data sets' complexity, which is associated with the neural network models' accuracy in forecasting.

Project Advisor — June 2023 – Present
Project Advisor at Nanjing Forestry University — China
- Provided technical support for COVID-19 epidemic trend Prediction models.
- Analyzed the one-on-one relationship among energy consumption, carbon emissions, GDP, and forest cover rate in China with both OLS-based and LSTM-based Deep Learning models.

Teaching Assistant
University of Rochester / URMC — Rochester, NY
- **CSC242: Artificial Intelligence** (CS Dept.) — Aug 2022 – Dec 2022
  Assisted in teaching adversarial search, Bayesian networks, and multilayer perceptrons (Java).

- **BST401: Probability Theory** (Dept. of Biostatistics) Aug 2025 – Dec 2025
  Assisted in teaching probability spaces, random variables, independence, distributions, expectation, convergence, LLN, and CLT.
- **BST426: Linear Models** (Dept. of Biostatistics) Jan 2026 – May 2026
  Assisted in teaching least squares, GLM theory, estimability, hypothesis tests, CIs, regression, ANOVA/ANCOVA, mixed effects, correlation, and prediction.

Schwartz Discover Scholar Showcase Oct 2023
poster fair for University of Rochester 2023 Schwartz Discover scholars Rochester, NY

- Presented the joint summer research group works with the poster entitled "A Novel Approach for Assessing the Predictability of Time-Series."

Research Paper and Preprints

Livia Betti, Ivan Chio, Julie Fleischman, Alex Iosevich, Filippo Iulianelli, Scott Kirila, Michele Martino, Azita Mayeli, Svetlana Pack, **Zhelin Sheng**, Conor Taliancic, Andrew M. Thomas, Nathan Whybra, Emmett Wyman, Ustun Yildirim, and Kaiyuan Zhao. "Fractal dimension, approximation, and data sets." In Combinatorial and Additive Number Theory VI. CANT CANT 2022 2023. Springer Proceedings in Mathematics & Statistics, vol 464. Springer, Cham. https://doi.org/10.1007/978-3-031-65064-2_3 (2025).

Zhelin Sheng, Kaimei Zhang, Chen Ling, Wenjuan Shen, Zihan Zhang, Chuanxin Ma, Changlei Xia, Keyi Chen, Yu Shen, Yu Hao, and Jiangang Han. "The forest carbon paradox: novel insights into China's forest-economy-emissions relationships." *npj Climate Action* (2026) 5:26. https://doi.org/10.1038/s44168-026-00350-w (2026).

Awards and Honors

Dean's List
University of Rochester 2021 – Present

SOUTHBURY MIDDLEBURY SCHOLARSHIP
SMSF grants, $3000 per year 2020 & 2022

Conference and Seminars attended

ag-conference Topics in microlocal analysis, harmonic analysis, and inverse problems, August 15-17, 2022

Skills

Programming Languages: Python, Java, R, Kotlin, JavaScript, HTML/CSS, PHP
Tools and Framework: LaTeX, Tensorflow, Pytorch

Sample course work

Including ongoing course: Discrete Mathematics, The Mathematical Experience (Putnam competition training course), Intro to Probability, Numerical Analysis, Intro to Complex Var With App (Complex Analysis), Functions of Real Variable (Real Analysis),
Combinatorial Analysis, Honors sequence Math: Honors Calculus I, II, III and IV,

Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Computer Models of Perception & Cognition, Design & Analysis Efficient Algorithms.